

# Aditi Reddy Doma

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## EDUCATION

### Masters of Science in Data Science

Rochester Institute of Technology (GPA: 4.0/4.0)

**Expected: December 2026**

Rochester, New York

**Relevant Coursework:** Applied Statistics(A), Database Management(A), Visual Analytics(A), Software Construction(A)

### Bachelor of Technology in Artificial Intelligence

Mahindra University (GPA: 3.56/4.0)

**July 2020 - May 2024**

Hyderabad, India

**Relevant Coursework:** Machine Learning(A-), Data Structures & Algorithms(A), Generative Artificial Intelligence(A-), Operating Systems(A-)

## SKILLS

- **Programming Languages:** Python, Java, R, SQL
- **Machine Learning:** Pandas, NumPy, Scikit-Learn, Keras, TensorFlow, Jupyterhub, PyTorch
- **Cloud Platforms:** AWS, Kafka, Databricks, Apache Spark
- **Tools & Technologies:** Tableau, Power BI, Looker, MS Excel, MS Word, MS PowerPoint, Statistics, Minitab, ETL, Git

## WORK EXPERIENCE

### Data Science Researcher

iCitizen

**May 2025 - Present**

Rochester, NY

- **Web scraped and analyzed 500K+ unstructured congressional profiles** from Congress.gov and Bioguide websites using APIs, transforming unstructured data into structured datasets stored in **AWS DynamoDB** for **real-time updates**.
- **Designed a dynamic U.S. map** using **D3.js**, visualizing senators, their sponsored bills and co-sponsors by state.
- **Increased user engagement by 27.2%** on the iCitizen app by **developing a chatbot** predicting likelihood of a bill becoming law.

### Graduate Teaching Assistant

Rochester Institute of Technology

**January 2025 - Present**

Rochester, New York

- Guided 200 undergraduates in **database systems** and **SQL** through labs focused on **query writing** and **relational data modeling**.

### Data Analyst

StandardWings Technologies Pvt. Ltd.

**August 2023 - August 2024**

Nashik, India

- Extracted, cleaned and integrated **multi-source data** (GPS, sensor, Salesforce CRM, SAP, Excel) from 120+ vehicles into **AWS S3** using **SQL** and **Python**, enabling detailed fleet analysis that uncovered significant operational cost savings.
- Collaborated with **cross-functional teams** throughout the project lifecycle to define **21 key KPIs** aligned with business objectives.

### Machine Learning Intern

Cloud4C

**June 2023 - July 2023**

Hyderabad, India

- Achieved **over 90% accuracy** in a **sentiment analysis model** using **BERT** to analyze customer feedback.
- **Performed large-scale text analysis** on customer feedback using **Databricks** with **PySpark** for data processing.
- Identified **top 5 recurring customer concerns** using **unsupervised learning (K-Means clustering)** on BERT embeddings.

## PROJECTS

### Portfolio Allocation Model ([GitHub](#))

**September 2022 - December 2022**

- Researched and scraped **5 years of stock data** for 9 companies across 3 sectors using Yahoo Finance and optimized portfolio allocation through **mean-variance modeling** using **Pyomo**, **NumPy** and **Pandas**.
- Identified Buy & Hold as the optimal investment strategy with **15% ROI** by performing **statistical analysis** of risk-return trade-off assessments using **Monte Carlo simulation** in **Python**.

### Covid Data Exploration and Visualization ([GitHub](#))

**August 2023 - October 2023**

- **Identified COVID-19 and vaccination impact trends** using **MS Excel** and **SQL queries**. Visualized the trends in **Tableau**.

### Prime Video Dashboard ([GitHub](#))

**September 2024 - November 2024**

- Designed and deployed an interactive **Power BI dashboard** to monitor and visualize Prime Video's content distribution, analyzing viewer demographics, content ratings and genre trends.
- Enabled real-time **data exploration** and **decision-making** through advanced analytics, facilitating detailed user behavior insights.

### Brand Logo Classification ([GitHub](#))

**February 2025 - April 2025**

- **Reduced classification time by 60%** and **improved accuracy by 20%** by implementing a **Convolution Neural Network (CNN)** classification system with **transfer learning** capabilities to categorize brand logos.
- Achieved **92.17% accuracy** with **VGG16** through **hyperparameter tuning**, **early stopping** and implementation of **Adam optimizer**.

### Energy Consumption Predictor ([GitHub](#))

**June 2025 - August 2025**

- Forecasted hourly energy consumption with **92.54% accuracy** using **Supervised Learning (Linear Regression)** in **R** for **5K+ homes**.
- Designed a **Shiny dashboard** to visualize how temperature, time and house attributes **influence future energy consumption**

### IOT Telemetry Analytics for Health Monitoring Devices ([GitHub](#))

**September 2025 - December 2025**

- Developed **ETL pipeline** using **Kafka** + **PySpark** to process **2M+ IoT telemetry records** from health devices (glucose, heart-rate).
- **Reduced false alarms by 18%** by **fine-tuning anomaly detection models (Isolation Forest, ARIMA)** for abnormal health data.